

favorable risk/reward opportunities. Even simple models can be profitable if they slightly increase your odds of success and are combined with proper risk management.

Combining Techniques: A Holistic Analytical Approach

The most powerful insights often come from combining multiple analytical techniques. Let's look at how professional traders might analyze a stock using several methods together:

```
1. # A more comprehensive analysis approach
2. import yfinance as yf
3. import pandas as pd
4. import numpy as np
5. import matplotlib.pyplot as plt
6. import seaborn as sns
7. from statsmodels.tsa.stattools import adfuller
8. from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
9.
10. # Get stock data
11. ticker = 'AAPL'
12. data = yf.download(ticker, start='2020-01-01', end='2023-01-01')
13.
14. # Calculate returns and volatility
15. data['Returns'] = data['Adj Close'].pct_change()
16. data['Volatility'] = data['Returns'].rolling(window=21).std() * np.sqrt(252) * 100 # Annualized
17.
18. # 1. Summary statistics of returns
19. print("Summary Statistics:")
20. print(data['Returns'].describe())
21.
22. # 2. Create visualization dashboard
23. plt.figure(figsize=(15, 12))
24.
25. # Price and volume chart
26. plt.subplot(3, 2, 1)
```